

应用案例

APPLICATION CASES

1. 精准医疗普惠化

Universal Access To Precision Medicine

标准化的操作, 确保治疗效果的一致性, 降低医生经验依赖性, 延长医生职业寿命。

基层医院借力机器人实现三甲级手术精度, 缓解医疗资源分布差距, 提升医疗资源可及性。

Standardized protocols ensure consistent therapeutic outcomes, minimize dependence on individual expertise, and extend surgeons' professional careers

Empowered by robotic-assisted systems, primary hospitals achieve top-level hospital surgical precision, bridging healthcare resource gaps and advancing equitable care

2. 多模态风险防控

Multimodal Risk Control

实时监测手术, 提供精确的术中数据, 及时纠正偏差, 减少手术风险。

通过可视化和多模态反馈 (触觉、声音、图像), 全面掌控手术进展。

Real-time intraoperative monitoring provides accurate surgical data, promptly corrects deviations, and reduces procedural risks

Achieve comprehensive intraoperative mastery through visualization and multimodal feedback (haptic, auditory, visual), facilitating real-time optimization of surgical trajectories

3. 智能医疗新篇章

A New Era in Intelligent Healthcare

整合人工智能和大数据分析, 为个性化医疗提供支持, 优化手术方案。

手术数据沉淀以及5G技术的发展, 为远程手术、数字骨科奠定技术基座。

Integrating artificial intelligence and big data analysis to support personalized medicine and optimize surgical plans

The accumulation of surgical data and the advancement of 5G technology lay the technical foundation for remote surgery and digital orthopaedics.



秉持创新精神,
用前沿科技打造高端医疗智能装备,
帮助医生造福患者

Adhering to the spirit of innovation,
we use cutting-edge technology to create high-end
medical intelligent equipment
to help doctors benefit patients.

北京威高智慧科技有限公司
Beijing Weigao Technology Co.,Ltd

(010) 6280-0509 weiztechnology@weiz.cn

国际医疗器械城 (东区) 二期 6号楼三层西侧
International Medical Equipment City (East Zone), Phase II,
Building 6, 3rd Floor, West Side

WEGO 威高



TOTAL KNEE
ARTHROPLASTY NAVIGATION
AND POSITIONING SYSTEM

全膝关节置换手术导航定位系统

注册证号: 国械注准20243012468

产品特点
PRODUCT FEATURES



• 个性化定制
Personalized Customization

Ct影像处理, 结合患者实际, 智能设计植入方案, 使手术更加胸有成竹。

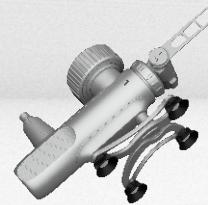
Automatic generation of 3D anatomical models based on CT data with deep learning algorithms and consequent implant positioning optimization, ensuring predictable surgical outcomes.



• 实时动态追踪
Real-time Dynamic Tracking

全程洞悉关节动态变化, 实时间隙平衡反馈, 随时方案调整, 实现最理想的下肢对线和软组织平衡。

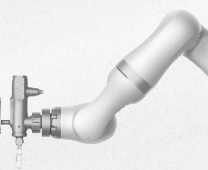
Continuous monitoring of intraoperative joint kinematics and ligament tension, providing instant feedback for surgical plan adjustments to achieve optimal lower limb alignment and soft tissue equilibrium.



• 安全保护
Real-Time Safety Boundary Enforcement

根据术前规划, 设定截骨安全区域, 避开血管及神经, 降低软组织损伤风险。

Virtual haptic boundaries aligned with preoperative plans to restrict robotic tool motion, minimizing risks of neurovascular injuries.



• 精准执行
Precision-enforced Execution

全程可视化截骨, 实时反馈截骨效果, 具备患者随动功能, 保证截骨精度, 提高手术成功率, 减少并发症。

Bone resection accuracy is ensured within $\pm 0.5\text{mm}$ by combining optical navigation with real-time motion compensation and continuous depth/angle monitoring.

高精度光学相机
High-Precision Optical Camera



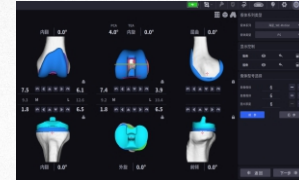
智能手术控制平台
Intelligent Surgical Control Platform



机械臂执行系统
Robotic Arm Actuation System



操作流程
OPERATIONAL PROCESS



1、术前获取影像资料, 制定手术方案
Generates patient-specific 3D surgical plans, including implant sizing and component positioning.



2、术中通过配准, 使虚实保持一致
Achieves spatial matching between preoperative 3D models and intraoperative anatomy through fiducial marker-based registration.



3、通过动态追踪, 使间隙平衡
Quantifies joint space variations during range-of-motion testing and dynamically adjusts osteotomy parameters for gap balance.



4、机械臂走位, 末端动力截骨
Executes predefined surgical plan by collaborating with the robotic arm equipped with various orthopedic power systems.



5、医生完成假体安装
Surgeon verifies implant fitting and completes its installation.